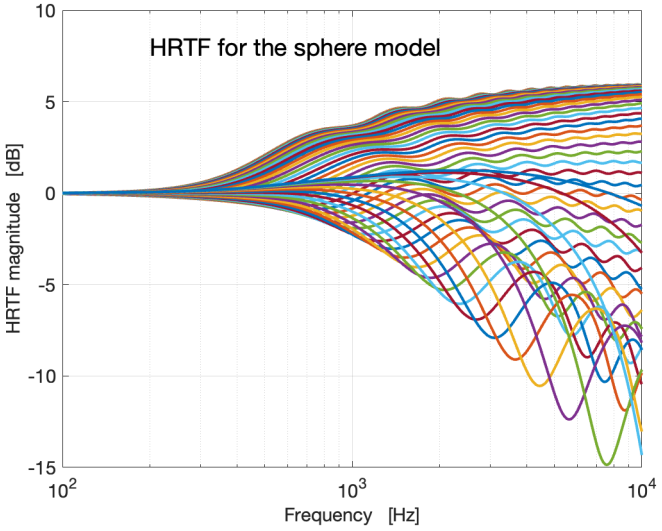
Assignment 2 - Directional hearing

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Your deliverable should include the following:

- One plot of an HRIR-pair (impulse responses) where you have implemented the IIR-filter and the interaural time delay (ITD). You should choose an incidence angle so that the ITD is clearly visible.
- One sound file of noise bursts moving around the head.
- A plot of the frequency response of your occlusion filter, and one sound example demonstrating occlusion (Task 4).
- A short written listening reflection and a short discussion of the model's limitations (Task 5).

If you get this assignment for your report, then all tasks must be included in the report.

Introduction

In this assignment, you should generate some sound examples using the technique called *binaural synthesis*. This technique exploits the properties of humans directional hearing. Central to the binaural synthesis is the concept of *Head-Related Transfer Functions (HRTFs)*. The principles for binaural synthesis are quite simple and elegant [1]:

- Our hearing has two input channels (we ignore vibrations in our bodies for now): the two sound pressure signals at the two eardrums. Therefore, if we *record* the two sound pressure signals at the eardrums, and reproduce them at a later instant, we should hear the same thing as at the original recording event. This is the *binaural recording technique* [2].
- As a consequence of the bullet point above, if we can *synthesize* the two eardrum sound pressure signals, for a real or virtual situation, then the listening person should get the same listening experience as in the simulated situation. This is the *binaural synthesis technique*.
- So, what is the sound pressure at the eardrum? That is described by the *Head-Related Transfer Functions (HRTFs)*, which is the TF of a filter, the output of which is the sound pressure at the eardrum, and the input signal of which is the sound pressure in an incoming sound wave (before it hits a listener's head). There will be one HRTF for each incidence angle, and one for each ear[1].

After the initial euphoria over the remarkably simple principle to generate 3D sound, one realizes that there are some challenges to solve:

1. The HRTFs will depend on the shape and size of a person's head, shoulders and outer ear properties. Thus the HRTFs are individual.
2. The HRTFs will be needed for a large number of incidence angles, since we can detect incidence angle differences down to 1 degree (in the most sensitive direction).
3. If we use headphones to generate the sound at the eardrums, we usually need to equalize the headphone's frequency response, to make sure that the sound pressure which arrives to the eardrum is as intended.
4. If we use headphones to generate the sound at the eardrums, we might need to use head-tracking and dynamically update the HRTFs used, to take into account that a listener deliberately and involuntarily moves the head more or less.
5. The ultimate input signal is the sound pressure at the eardrum of the listener, but it is very impractical to have to record at the eardrum (for capturing the sound, and to measure/check that the reproduced sound

[1] If the sound source is closer than approx. 1 m, the HRTFs will also be distance-dependent.

is as intended). A more common reference sound pressure is the sound pressure at the entrance to a blocked ear canal. With this choice, it turns out that the individual differences between HRTFs will be smaller than if the sound pressure at the eardrum is recorded.

HRTF alternatives

It is possible to choose more or less realistic HRTFs, which correspond to more or less complicated processing (for acquiring the HRTFs and/or for doing the filtering). Four levels of realism are:

1. The most accurate way is to use *individual HRTFs* for each listening person. It is, however, complicated to measure the HRTFs in an accurate way.

2. A much simpler approach is to represent the listening person's head with a *rigid sphere in free air*, that is, without any torso or ear pinna (outer ears). The sound pressure at two points on the sphere can then represent the sound pressure which enters the ear canal. Importantly, these two sound pressures would have realistic magnitudes and phases (the phase of the two signals represents the interaural time difference). However, the mathematical expressions for the sound pressure at a rigid sphere are somewhat complicated.

3. An even simpler approach than the rigid-sphere model is to use a *first-order IIR filter* for the magnitude of the (simplified) HRTFs, *combined with FIR-filters* to generate the correct interaural time delay, as suggested by Brown and Duda [3].

4. The simplest approach is to only use interaural time differences (ITD) as HRTFs. This gives a completely unrealistic spectrum, and no interaural level differences, but the important ITD will be correct.

Brown and Duda's simplified HRTFs

A first-order filter that modifies the magnitude for the "sunny-side ear" as well as the "shadow-side ear" was suggested by C. P. Brown and R. O. Duda in "An efficient HRTF model for 3D-sound," in Proc. of the IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), 1997. They suggested a filter response which can act as anything between a *shelving filter* which boosts high frequencies, to a low-pass filter:

$$H_{L/R}(\omega) = \frac{\alpha_{L/R} \cdot j\omega + \beta}{j\omega + \beta} \tag{1}$$

where

$$\beta = \frac{2c}{a},$$

a = the radius of a sphere that represents the head, c = the speed of sound, and

$$\alpha_L = 1 - \sin\theta, \qquad \alpha_R = 1 + \sin\theta, \tag{2}$$

where θ is the incidence angle *relative to the frontal direction*.

Thus, if the sound is coming from straight ahead,

$$\theta = 0 \Rightarrow \begin{cases} \alpha_L = 1 \\ \alpha_R = 1 \end{cases}$$

If the sound is coming from the right side,

$$\theta = \frac{\pi}{2} = 90^\circ \Rightarrow \begin{cases} \alpha_L = 0 \\ \alpha_R = 2 \end{cases}$$

and if the sound is coming from the left side,

$$\theta = -\frac{\pi}{2} = -90^\circ \Rightarrow \begin{cases} \alpha_L = 2 \\ \alpha_R = 0 \end{cases}$$

etc for other angles $\theta \in [-90^\circ, 90^\circ]$.

How can we construct a digital filter from the frequency response expression above? The bilinear transform converts a Laplace transform, $H(s)$, to a z-transform, $H(z)$, via:

$$s = \frac{2}{T} \cdot \frac{(z-1)}{(z+1)} \tag{3}$$

where $T = 1/f_S$. Our given frequency response on the $j\omega$-form can be converted to the Laplace transform via $s = j\omega$:

$$H_{L/R}(s) = \frac{\alpha_{L/R} \cdot s + \beta}{s + \beta}$$

Using the bilinear transform above, you can get $H_{L/R}(z)$, and if you write it on this form,

$$H_{L/R}(z) = \frac{B_0 + B_1 z^{-1} + B_2 z^{-2} + ...}{1 + A_1 z^{-1} + A_2 z^{-2} + ...} \tag{4}$$

you have IIR-filter coefficients directly!

The interaural time difference can be handled separately with an FIR filter with a delayed pulse: all frequencies will be delayed with the same amount. The delay can be found using the simple formula called Woodworth's formula:

$$\Delta t = \frac{a}{c}(\theta + \sin\theta) \tag{5}$$

where θ is the angle relative to straight ahead, that is, the same as in Eqs. (2). Note that you get positive and negative delays for positive and negative θ with Eq. (5) - make sure that the shadow side of the head gets the sound later than the sunny side!

Tasks

In all your tasks, the sampling frequency should preferably be 44100 Hz. For the speed of sound you could choose 343 m/s, and for the head radius you could choose $a = 9$ cm.

1. **The simple HRTF filter - in the analog and digital domains.**

   The expression in Eq. (1) is the analog frequency-domain expression suggested by Brown and Duda. It can be transformed into a first-order IIR filter using the bilinear transform given by Eq. (3). In this task, you should:

   (a) Derive the IIR filter coefficients B_0, B_1, A_1 by transforming the expression in Eq. (1) using the bilinear transform in Eq. (3).

   (b) Derive the frequency response of the digital IIR filter by replacing each delay, z^{-1},by $\mathrm{e}^{-j\omega T} = \mathrm{e}^{-j\omega/f_S}$.

   (c) Plot the frequency response given by Eq. (1) and the response you have derived from the IIR filter. Do this for a few incidence angles, and plot both the left-ear and right-ear responses in the same diagram.

2. **The simple IIR filter combined with the ITD**

   The IIR filter in the previous task gives the right frequency response magnitude, but not the ITD. You should combine the result in the previous task with ITD-s to generate a complete HRIR simulator!

   Eq. (5) gave an expression for the delay to the right and left ear, which can be converted to an integer number of samples delay.

   The delay to the "sunny-side ear" will be negative, which means the impulse response will start before the time zero (it is non-causal). This can easily be fixed by adding the same constant number of samples to both the left and right ear impulse responses!

   Your tasks are:

   (a) Give an expression for the ITD from the two Δt in Eq. (5) *after rounding to an integer number of samples.*

   (b) Plot the ITD based on the exact expression in Eq.(5), and for the rounded expression in the previous sub-task, as function of incidence angle θ.

(c) Plot a few examples of impulse reponses where you have combined the IIR filter in the previous task and the delays in this task. Choose a few incidence angles so one can see the ITD.

3. The final sound demonstrations

(a) Demonstration 1
Generate a sound example where you take a sound signal (music, noise, or whatever you prefer) and simulate short bursts that arrive from different incidence angles. It is quite effectful to let short noise bursts move around your head in steps of e.g., 30 degrees.

Hint: it is preferable to use pink noise bursts rather than white noise bursts, because pink noise bursts are perceived as having a more even spectrum than white noise (which is perceived as sharp, or high-frequency dominated).

Hint 2: Beware of the filter *function in Matlab. If you use this function for filtering with an FIR filter, please note that* the output signal is the same length as the input signal*. A convolution with an FIR filter should give an output signal which is longer than the input signal, and the effective truncation caused by the* filter *function might or might not have an unwanted effect for you. It is, however, very easy to avoid any potential problems by making sure that all input signals that you will filter with will have a number of zeros at the end which is larger than the length of the FIR filters.*

(b) Demonstration 2
Generate a binaural impulse response for the scenario that a sound source is 2m in front of you (the listener), **and there is a reflection from the same direction as the sound source which gives a 0.5m longer path than the direct sound**. Assume that the sound source is omnidirectional and assume that the wall is perfectly reflecting. Convolve this binaural impulse response with a sound signal (music, noise, or whatever you prefer).

Hint: You can assume a spherically spreading woundwave from the sound source, which means that the amplitude is proportional to 1/r where r is the distance from source to receiver.

(c) Demonstration 3
Generate a binaural impulse response for the scenario that a sound source is 2m in front of you (the listener), **and there is a reflection from the direction 90 degrees from the left, 60 ms after the direct sound**. Scale this impulse response so that it corresponds to a spherical wave from the source, giving a direct sound at 2m distance, and a reflection from the side arriving 60 ms later, being reflected from a perfectly reflecting wall. Convolve this binaural impulse response with a **transient** sound signal (e.g., a hand clap).

4. Simulating occlusion

When an obstacle (e.g., a wall, a piece of furniture, another person) partially blocks the direct path between the source and the listener, the sound that reaches the listener is not simply attenuated uniformly across frequency. Because low frequencies diffract around edges more easily than high frequencies, an occluded sound is perceived as "muffled": high frequencies are attenuated much more than low frequencies — the same effect you hear through an almost-closed door, where the bass comes through but the higher frequencies are lost.

A simple way to approximate this is with a first-order low-pass filter, in the same analog form as the *shelving filter* in Eq. 1, but with $\alpha = 1$ (no high-frequency boost) and with β chosen to represent the occlusion cutoff frequency:

$$H_{occ}(s) = \beta_{occ}/(s + \beta_{occ}), \beta_{occ} = 2\pi f_c \tag{6}$$

where f_c is the cutoff frequency (in Hz) you choose to represent how "occluded" the sound is (a lower f_c represents a more strongly occluded/muffled sound).

Your tasks are:

(a) Using the same bilinear-transform approach as in Task 1, derive the IIR filter coefficients for $H_{occ}(z)$.

(b) Plot the frequency response of $H_{occ}(z)$ for a couple of different cutoff frequencies, e.g. $f_c = 500$ Hz and $f_c = 2000$ Hz.

(c) Apply this filter to the source signal before the HRTF/ITD processing in one of your demonstrations from Task 3 (e.g., regenerate Demonstration 2 with the direct sound occluded), and briefly describe, in your report, what audible difference the occlusion filter makes.

Hint: this is a simplification — real diffraction around an obstacle depends on the obstacle's size relative to the wavelength in a much more complex way. This first-order filter is only a coarse approximation, in the same spirit as the simplified HRTF model used throughout this assignment.

5. Reflection questions

(a) *Listening reflection:* Listen to your sound examples from Task 3 and Task 4 over headphones. In a few sentences, describe what you actually perceive: does the sound genuinely seem to move around your head in Demonstration 1? Is the effect equally convincing in all directions, or stronger or weaker for some incidence angles than others? Does the reflection in Demonstrations 2 and 3 change your

perception of the spectral quality of the sound? Do you hear any echo effect? Do you perceive any direction to the sound(s)?

(b) *Model limitations:* The HRTF model you have implemented (Brown and Duda's simplified model) is one of the simplest of the four alternatives described in the Introduction. Based on the assumptions behind this simplified model, name and briefly explain two specific listening cues or effects that this model cannot reproduce correctly, and explain why.

References

[1] P. Svensson, "TTT4295 Acoustic Signal Processing," Dept. of Electronic Systems, NTNU, Trondheim, 2024.

[2] H. Møller, "Fundamentals of Binaural Technology," Appl. Acoust. 36, pp. 171-218 (1992).

[3] C. P. Brown and R. O. Duda, "An efficient HRTF model for 3D-sound," in Proc. of the IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), 1997.